

TUTTE EVALUATIONS ON METRIC GRAPHS AND RANK-GENERATING POWER SERIES

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ABSTRACT. We define a formal power series attached to any metric graph with arbitrary positive real edge lengths, which recovers the usual Tutte polynomial when all edge lengths are positive integers. We prove that evaluation of the rank-generating power series defines a continuous function of the moduli space of metric graphs, under certain constraints on the input parameters. The evaluations give rise to a large family of continuous functions generalizing the one that sends a metric graph to the volume of its graph Jacobian. This is related to string theory.

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1. INTRODUCTION

Given a graph G , the Tutte polynomial $T(G; x, y)$ is a two-variable polynomial introduced by Tutte [21]. Many important graph invariants arise as evaluations of the Tutte polynomial at specific integer parameters x, y . In particular, $T(G; 1, 1)$ is the number of spanning trees of G . In tropical geometry, it is well-known that the number of spanning trees on a graph has a “continuous generalization” as the volume of the Jacobian of a metric graph (a.k.a. an abstract tropical curve) [13, 3, 2]. This motivates the question: can other Tutte polynomial evaluations be generalized to arbitrary metric graphs? Other Tutte polynomial evaluations include counts of connected subgraphs, acyclic subgraphs, acyclic orientations, and totally cyclic orientations. In this paper, we show that the function sending a graph G to its Tutte polynomial evaluation $T(G; x, y)$ can be extended to a continuous function on the space of metric graphs, whenever $x > 0$. This is achieved by defining a rank-generating power series on a metric graph, which generalizes the usual one on a graph.

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As a consequence, when $x > 0$, our definition of Tutte evaluation at a chosen (x, y) gives a continuous function on the space of metric graphs of fixed genus.

Theorem 1. *For fixed real constants x and y , let $\text{ev}_{(x,y)}(G)$ denote the evaluation of the Tutte polynomial of G at (x, y) . If $x > 0$, then $\text{ev}_{(x,y)} : \mathcal{M}_g^{\text{graph}} \rightarrow \mathbb{R}$ defines a continuous function on the space of metric graphs of genus g .*

fix theorem statement

For a comprehensive modern overview of the Tutte polynomial see [6, 9].

1.1. Rank-generating power series. The following characterization of the Tutte polynomial was initially introduced by Crapo [8], using the *rank-generating function* of G (see also [9, Definition 3]). Given a connected graph G , the Tutte polynomial of G is

$$\text{eq:tutte-graph} \quad (1) \quad T(G; x, y) = \sum_{A \subseteq E(G)} (x-1)^{h_0(G \setminus A)-1} (y-1)^{h_1(G \setminus A)}$$

where $G \setminus A$ denotes the graph with edges in A deleted, and h_0 and h_1 denote the zeroth and first Betti numbers of a topological space. In graph theoretic terms,

$$\begin{aligned} h_0(G) &= \#(\text{connected components of } G), \quad \text{and} \\ h_1(G) &= \#E(G) - \#V(G) + h_0(G). \end{aligned}$$

Suppose Γ is a metric graph with combinatorial model $\Gamma = (G, \ell)$, where $\ell : E(G) \rightarrow \mathbb{R}_{>0}$ assigns a positive length to each edge of G .

We define the *rank-generating power series* $T^+(\Gamma; u, w)$ of Γ as

$$\text{eq:tutte-power-series} \quad (2) \quad T^+(\Gamma; u, w) = \sum_{A \subseteq E(G)} \left(\prod_{e_i \in A} \llbracket \ell(e_i) \rrbracket_{1+u} \right) u^{h_0(G \setminus A)-1} w^{h_1(G \setminus A)}$$

where $\llbracket \alpha \rrbracket_{1+u}$ denotes the formal power series

$$\text{eq:f-analog} \quad (3) \quad \llbracket \alpha \rrbracket_{1+u} := \alpha + \binom{\alpha}{2} u + \binom{\alpha}{3} u^2 + \cdots \in \mathbb{R}[[u]]$$

and $\binom{\alpha}{k}$ denotes the polynomial $\frac{1}{k!} \alpha(\alpha-1) \cdots (\alpha-k+1)$. If all edges in Γ have unit length, i.e. $\ell(e) = 1$, then (2) recovers the Tutte polynomial by the substitution $T(G; x, y) = T^+(\Gamma; x-1, y-1)$.

1.2. Statement of results. Our first main result is that the expression (2) does not depend on the chosen model (G, ℓ) for the metric graph Γ .

Theorem 2 (Rank-generating power series). *Given a metric graph $\Gamma = (G, \ell)$, the expression $T^+(\Gamma; u, w)$ is a well-defined power series in $\mathbb{R}[[u]][w]$; in particular, $T^+(\Gamma; u, w)$ does not depend on the choice of model (G, ℓ) for Γ .*

To prove this result, we show that $T^+(\Gamma; u, w)$ satisfies a familiar deletion-contraction identity (Theorem 13).

1.3. Evaluating rank-generating power series. Instead of considering a fixed metric graph Γ , we can vary Γ over a natural moduli space of metric graphs. As Γ varies, the value of $T^+(\Gamma; u, w)$ also varies continuously under certain mild assumptions on the parameters (u, w) .

Let $\mathcal{M}_g^{\text{graph}}$ denote the moduli space of stable metric graphs of genus g .

Theorem 3 (Continuity of Tutte evaluation). ^{thm:tutte-eval-moduli} Let u and w be fixed real numbers, with $u > -1$. The Tutte evaluation at (u, w) ,

$$\text{ev}^+(u, w) : \Gamma \mapsto T^+(\Gamma; u, w),$$

defines a continuous function $\text{ev}^+(u, w) : \mathcal{M}_g^{\text{graph}} \rightarrow \mathbb{R}$ on the moduli space of stable graphs. Namely,

- (i) (Continuity on cells) for each stable combinatorial graph G , the Tutte evaluation $\text{ev}^+(u, w)$ defines a continuous function

$$\text{ev}^+(u, w) : \mathbb{R}_{>0}^{E(G)} \rightarrow \mathbb{R},$$

where a point in the positive orthant $\mathbb{R}_{>0}^{E(G)}$ represents a choice of positive, real edge lengths in G .

- (ii) (Continuity between cells) As the length of a non-loop edge $e \in E(G)$ approaches zero in the metric graph $\Gamma = (G, \ell)$ while all other edge lengths are fixed, the value of $\text{ev}^+(u, w)$ at (G, ℓ) approaches the value of $\text{ev}^+(u, w)$ at the contraction Γ/e .

Note the power series (3) converges for $|u| < 1$, and in this range satisfies

$$\llbracket \alpha \rrbracket_1 = \alpha \quad \text{and} \quad \llbracket \alpha \rrbracket_{1+u} = \frac{(1+u)^\alpha - 1}{u} \quad \text{if } u \neq 0.$$

Example 4 ($u = 0, w = 0$). On a metric graph, the Tutte evaluation $\text{ev}^+(0, 0)$ gives the volume of the Jacobian of $\Gamma = (G, \ell)$, which can be expressed as a weighted sum of spanning trees of G . The function $\text{ev}^+(0, 0)$ is continuous on $\mathcal{M}_g^{\text{graph}}$, and extends continuously to $\overline{\mathcal{M}_g^{\text{graph}}}$ (where it has value zero on the boundary). [cite a reference]

Example 5 ($u = -1, w = 1$). The Tutte evaluation $\text{ev}^+(-1, 1)$ on $\Gamma = (G, \ell)$ gives the number of totally cyclic orientations of Γ . This number does not depend on the edge lengths of Γ ; i.e. $\text{ev}^+(-1, 1)$ is constant on metric graphs of a fixed combinatorial model G . However, $\text{ev}^+(-1, 1)$ is not continuous as some edge length approaches 0. Namely, the value of $T(G; -1, 1)$ generally differs from the value of $T(G/e; -1, 1)$ after contracting an edge.

For example, the number of totally cyclic orientations of graphs of genus two are shown below.

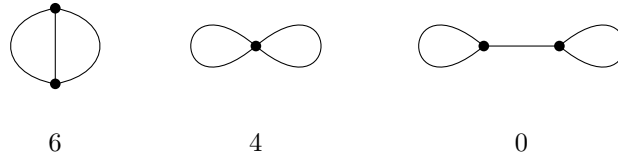


FIGURE 1. Number of totally cyclic orientations.

1.4. Previous work. The multivariate Tutte polynomial [17] is also known as the *Potts-model partition function*. Sokal [17] asks:

Let me conclude by observing that numerous specific evaluations of the Tutte polynomial have been given combinatorial interpretations, as counting some set of objects associated to the graphs G . It would be an interesting project to seek to extend these counting problems to “counting with weights,” i.e., to obtain suitably defined univariate or multivariate generating polynomials for the objects in question as specializations of $Z_G(q, v)$ or $Z_G(q, \mathbf{v})$, respectively.

Is the “continuous” version of $Z_G(q, v)$ described? if so, link it

Several authors have investigated the behavior of the Tutte polynomial under the operation of subdividing an edge into multiple edges.

Previous work on *chain polynomials*:

The identities presented in this paper are essentially in the work of Read and Whitehead [16], but with the restriction that edge lengths are positive integers. In their work, edge lengths are called “chain lengths” since they are viewed as a chain of unit-length edges.

The family of graphs obtained by varying chain lengths of a fixed graph is called a “homeomorphism class” of graphs.

Read and Whitehead [16] [15, 16]

The essential contribution of this work is to enforce a dependence of the edge weights $(\alpha(e))$ on the polynomial parameter u , namely $\alpha(e) = \llbracket \ell(e) \rrbracket_{1+u}$.

Traldi [18] considers the weighted Tutte polynomial

$$\tilde{T}^+(G; u, w) = \sum_{A \subseteq E} \left(\prod_{e \in A} c(e) \right) u^{h_0(G|A)} w^{h_1(G|A)}.$$

The same weighted polynomial was previously studied by Fortuin–Kasteleyn [10], but in harder-to-understand notation for a modern reader.

Traldi [18, 19, 20]

Application: Zeros of Tutte polynomials? For example, [5] and [4] study zeros of chromatic polynomials in a family of graphs consisting of assigning integer-valued edge lengths to an underlying graph model.

Jackson and Sokal [11] study zero-free regions of the Tutte polynomial.

Ok and Perrett [14] study the density of the real zeros of the Tutte polynomial.

Merino–Welsh conjecture [12]: for any graph G , Merino and Welsh conjectured that $\max(T(G; 2, 0), T(G; 0, 2)) \geq T(G; 1, 1)$.

1.5. Questions. Do the coefficients of the rank generating power series $T^+(G; u, w)$ produce interesting metric graph invariants? Are these coefficients continuous on the space of metric graphs?

1.6. Notation. Γ a compact metric graph

G a finite graph, loops and parallel edges allowed, possibly disconnected

$E(G)$ edge set of G

$V(G)$ vertex set of G

(G, ℓ) a combinatorial model for a metric graph, where

$\ell : E(G) \rightarrow \mathbb{R}_{>0}$ is a length function on edges of G

$T(G; x, y)$ the Tutte polynomial of G

$T^+(G; u, w) = T(G; 1 + u, 1 + w)$ “additive” centered Tutte polynomial

$T^+(\Gamma; u, w)$ the rank-generating power series of Γ

2. q -ANALOGS

For a positive integer n , the q -analogue $[n]_q$ is defined as the polynomial

$$[n]_q = \frac{q^n - 1}{q - 1} = 1 + q + q^2 + \cdots + q^{n-1} \in \mathbb{Z}[q].$$

elaborate on these references below

When n is not an integer, $[n]_q$ does not admit a Laurent expansion in the variable q . However, we can obtain a well-defined power series under a simple change of variable. Namely, note that

$$[n]_{1+u} = \frac{(1+u)^n - 1}{u} = \sum_{k \geq 1} \binom{n}{k} u^{k-1},$$

and the binomial coefficient function $n \mapsto \binom{n}{k}$ has a well-defined extension

$$(4) \quad \binom{\alpha}{k} := \frac{\alpha(\alpha-1) \cdots (\alpha-k+1)}{k!}$$

where α is any real number.

Definition 6. Let $\llbracket \alpha \rrbracket_{1+u}$ denote the power series

$$(5) \quad \llbracket \alpha \rrbracket_{1+u} := \alpha + \binom{\alpha}{2} u + \binom{\alpha}{3} u^2 + \cdots \in \mathbb{R}[[u]].$$

TODO: decide on notation $\llbracket \alpha \rrbracket_q$ $[\alpha]_q$ $\llbracket \alpha \rrbracket_q$ $[\alpha]_q$

Fact 7.

- (1) For a positive integer n , the expression $[n]_q$ is a polynomial in $\mathbb{Z}[q]$. As $q \rightarrow 1$, we have $[n]_q \rightarrow n$. As $q \rightarrow 0$, we have $[n]_q \rightarrow 1$.
- (2) For a real number α , the expression $\llbracket \alpha \rrbracket_{1+u}$ is a power series in $\mathbb{R}[[u]]$. As $u \rightarrow 0$, we have $\llbracket \alpha \rrbracket_{1+u} \rightarrow \alpha$.

If α is positive, then as $u \rightarrow -1$, we have $\llbracket \alpha \rrbracket_{1+u} \rightarrow 1$.

For positive integers n, m we have

$$[n+m]_q = [n]_q + [m]_q + (q-1)[n]_q[m]_q.$$

The analogous identity holds for power series $\llbracket \alpha \rrbracket_{1+u}$.

prop:fanalog-add

Proposition 8. For real numbers α, β , we have

$$\llbracket \alpha + \beta \rrbracket_{1+u} = \llbracket \alpha \rrbracket_{1+u} + \llbracket \beta \rrbracket_{1+u} + u \llbracket \alpha \rrbracket_{1+u} \llbracket \beta \rrbracket_{1+u}.$$

as elements of $\mathbb{R}[[u]]$.

Proof. First, observe that on the open disc $|u| < 1$, we have

$$1 + u \llbracket \alpha \rrbracket_{1+u} = (1+u)^\alpha,$$

due to the binomial series identity. Thus

$$\begin{aligned} 1 + u \llbracket \alpha + \beta \rrbracket_{1+u} &= (1+u)^{\alpha+\beta} = (1+u)^\alpha (1+u)^\beta \\ &= (1 + u \llbracket \alpha \rrbracket_{1+u})(1 + u \llbracket \beta \rrbracket_{1+u}), \end{aligned}$$

which implies the desired result. □

3. METRIC GRAPHS

A metric graph is a compact, connected metric space which comes from assigning edge lengths to a finite, connected graph. If the metric graph Γ comes from a combinatorial graph G by assigning edge lengths $\ell : E(G) \rightarrow \mathbb{R}_{>0}$, we say (G, ℓ) is a *combinatorial model* for Γ and we write $\Gamma = (G, \ell)$.

3.1. Graph terminology.

$$\begin{aligned} h_0(G|A) - 1 &= rk(G) - rk(A), \quad \text{and} \\ h_1(G|A) &= \#(A) - rk(A). \end{aligned}$$

3.2. Tutte polynomial.

3.3. Deletion and contraction. Suppose $\Gamma = (G, \ell)$, and $e \in E(G)$ is a non-loop edge.

The *edge deletion* $\Gamma \setminus e$ is the metric graph obtained from Γ by removing the points in the interior of e . The *edge contraction* Γ/e is the metric graph obtained from Γ by removing the points in the interior of e , and then joining the endpoints of e .

4. RANK-GENERATING POWER SERIES

Let $T^+(\Gamma; u, w)$ be the power series in $\mathbb{R}[[u]][w]$ defined by

$$(6) \quad T^+(\Gamma; u, w) = \sum_{A \subset E(G)} \left(\prod_{e \in A} \llbracket \ell(e) \rrbracket_{1+u} \right) u^{h_0(G \setminus A) - 1} w^{h_1(G \setminus A)}.$$

Proposition 9.

(1) Let $\Gamma = \Gamma_1 \vee \Gamma_2$ denote the wedge sum of two metric graphs. Then

$$T^+(\Gamma_1 \vee \Gamma_2; u, w) = T^+(\Gamma_1; u, w) T^+(\Gamma_2; u, w).$$

(2) Let $\Gamma = \Gamma_1 \sqcup \Gamma_2$ denote the (disconnected) disjoint union of two metric graphs. Then

$$T^+(\Gamma_1 \sqcup \Gamma_2; u, w) = u T^+(\Gamma_1; u, w) T^+(\Gamma_2; u, w).$$

Example 10 (Rank-generating power series of a line). Suppose Γ is a segment of length α , then

$$T^+(\Gamma; u, w) = 1 + \alpha u + \binom{\alpha}{2} u^2 + \cdots.$$

If G is a line graph consisting of n edges, then

$$T^+(G; u, w) = (1 + u)^n = 1 + nu + \binom{n}{2} u^2 + \cdots + u^n.$$

Example 11 (Rank-generating power series of a circle). If Γ is a circle of length α , then

$$T^+(\Gamma; u, w) = w + \alpha + \binom{\alpha}{2} u + \binom{\alpha}{3} u^2 + \cdots.$$

Suppose G is a cycle graph consisting of n edges. Then

$$T^+(G; u, w) = n + \binom{n}{2} u + \binom{n}{3} u^2 + \cdots + nu^{n-2} + u^{n-1} + w = \frac{(1+u)^n - 1}{u} + w$$

Example 12 (Rank-generating power series of theta graph). Suppose G is the graph with two vertices connected by three edges. Suppose Γ is the metric graph which assigns lengths a, b, c to the edges of G . Then

$$\begin{aligned} T^+(G; u, w) &= (\llbracket a \rrbracket_{1+u} \llbracket b \rrbracket_{1+u} + \llbracket a \rrbracket_{1+u} \llbracket c \rrbracket_{1+u} + \llbracket b \rrbracket_{1+u} \llbracket c \rrbracket_{1+u}) \\ &\quad + (\llbracket a \rrbracket_{1+u} \llbracket b \rrbracket_{1+u} \llbracket c \rrbracket_{1+u}) u + (\llbracket a \rrbracket_{1+u} + \llbracket b \rrbracket_{1+u} + \llbracket c \rrbracket_{1+u}) w + w^2. \end{aligned}$$

5. DELETION-CONTRACTION

The Tutte polynomial $T(G; x, y)$ can be characterized inductively by the deletion-contraction relation:

$$T(G; x, y) = T(G \setminus e; x, y) + T(G/e; x, y)$$

if e is not a loop or bridge, along with the base cases

$$T(G; x, y) = x^i y^j \quad \text{if } G \text{ consists of } i \text{ bridges and } j \text{ loops.}$$

The rank-generating power series $T^+(\Gamma; u, w)$ satisfies a similar deletion-contraction relation.

Theorem 13 (Deletion-contraction relation). *Given a metric graph $\Gamma = (G, \ell)$ and an edge $e \in E(G)$, which is not a loop, the rank-generating power series satisfies*

$$(7) \quad T^+(\Gamma; u, w) = \llbracket \ell(e) \rrbracket_{1+u} T^+(\Gamma \setminus e; u, w) + T^+(\Gamma / e; u, w).$$

Proof. Observe that

$$T^+(\Gamma; u, w) = \sum_{\substack{A \subseteq E \\ e_0 \in A}} (\cdots) = \sum_{\substack{A \subseteq E \\ e_0 \in A}} (\cdots) + \sum_{\substack{A \subseteq E \\ e_0 \notin A}} (\cdots).$$

The first sum reduces to

$$\begin{aligned} & \sum_{\substack{A \subseteq E \\ e_0 \in A}} \left(\prod_{e \in A} \llbracket \ell(e) \rrbracket_{1+u} \right) u^{h_0(G \setminus A) - 1} w^{h_1(G \setminus A)} \\ &= \llbracket \ell(e_0) \rrbracket_{1+u} \sum_{A' \subseteq E \setminus e_0} \left(\prod_{e \in A'} \llbracket \ell(e) \rrbracket_{1+u} \right) u^{h_0(G \setminus e_0 \setminus A) - 1} w^{h_1(G \setminus e_0 \setminus A)} \\ &= \llbracket \ell(e_0) \rrbracket_{1+u} T^+(G \setminus e_0; u, w), \end{aligned}$$

while the second sum reduces to

$$\begin{aligned} & \sum_{\substack{A \subseteq E \\ e_0 \notin A}} \left(\prod_{e \in A} \llbracket \ell(e) \rrbracket_{1+u} \right) u^{h_0(G \setminus A) - 1} w^{h_1(G \setminus A)} \\ &= \sum_{A \subseteq E \setminus e_0} \left(\prod_{e \in A} \llbracket \ell(e) \rrbracket_{1+u} \right) u^{h_0(G \setminus A / e_0) - 1} w^{h_1(G \setminus A / e_0)} \\ &= T^+(G / e_0; u, w). \end{aligned}$$

In the second equality, we use the fact that the quotient map $(G \setminus A) \rightarrow (G \setminus A) / e_0$ is a homotopy equivalence, so it preserves homology groups. The third line uses the fact that deletion and contraction commute, $(G \setminus A) / e_0 = (G / e_0) \setminus A$. $\square$

5.1. Deleting bridges and contracting loops. In this section we describe how the definition of rank-generating power series $T^+(\Gamma; u, w)$ may be extended to a more general concept of metric graphs.

Definition 14. A *genus-weighted metric graph* $\Gamma = (G, \ell, \mathfrak{g})$ consists of a graph $G = (V, E)$, a length function $\ell : E \rightarrow \mathbb{R}_{>0}$ on edges, and a genus function $\mathfrak{g} : V \rightarrow \mathbb{Z}_{\geq 0}$ on vertices.

- If $\Gamma = \bigcup_{i=1}^k \Gamma_i$ is a disjoint union of k connected metric graphs Γ_i , then

$$T^+\left(\bigcup_{i=1}^k \Gamma_i; u, w\right) = u^{k-1} T^+(\vee_{i=1}^k \Gamma_i; u, w).$$

- If $\Gamma^{\mathfrak{g}} = (G, \ell, \mathfrak{g})$ is a genus-weighted metric graph, with underlying metric graph $\Gamma^0 = (G, \ell)$, then

$$T^+(\Gamma^{\mathfrak{g}}; u, w) = w^{\sum \mathfrak{g}(v)} T^+(\Gamma^0; u, w).$$

5.2. Proofs.

Proof of Theorem 2. It suffices to show that the rank-generating power series is invariant under an edge subdivision. Suppose $G = (V, E)$ contains the edge e_0 , which we subdivide into $e_1 \cup e_2$ to obtain the graph G' . Suppose the subdivided edges e_1 and e_2 have lengths a and b , so that e_0 has length $a + b$.

By the deletion-contraction relation, Theorem 13, we have

$$T^+(G; u, w) = \llbracket a + b \rrbracket_{1+u} T^+(G \setminus e_0; u, w) + T^+(G/e_0; u, w)$$

and

$$\begin{aligned} T^+(G'; u, w) &= \llbracket a \rrbracket_{1+u} T^+(G' \setminus e_1; u, w) + T^+(G'/e_1; u, w) \\ &= \llbracket a \rrbracket_{1+u} (\llbracket b \rrbracket_{1+u} T^+(G \setminus \{e_1, e_2\}; u, w) + T^+(G \setminus e_1/e_2; u, w)) \\ &\quad + (\llbracket b \rrbracket_{1+u} T^+(G/e_1 \setminus e_2; u, w) + T^+(G/\{e_1, e_2\}; u, w)) \end{aligned}$$

The compound contraction $G'/\{e_1, e_2\}$ is the same metric graph as G/e_0 , while the compound deletion $G' \setminus \{e_1, e_2\}$ is the disjoint union of $G \setminus e_0$ and an isolated vertex. For the mixed deletion-contraction operations, we have

$$G' \setminus e_1/e_2 \cong G'/e_1 \setminus e_2 \cong G \setminus e_0.$$

Therefore,

$$T^+(G'; u, w) = (\llbracket a \rrbracket_{1+u} + \llbracket b \rrbracket_{1+u} + u \llbracket a \rrbracket_{1+u} \llbracket b \rrbracket_{1+u}) T^+(G \setminus e_0; u, w) + T^+(G/e_0; u, w).$$

and from here it suffices to apply Proposition 8. $\square$

6. MODULI SPACES OF METRIC GRAPHS

See Melody Chan [7], and Abramovich–Caporaso–Payne [1].

A graph G is *stable* if it is connected, and every vertex has degree at least three.

The moduli space of (stable) genus g metric graphs is a finite polyhedral fan.

$$\mathcal{M}_g^{\text{graph}} = \bigsqcup \overline{C(G, w)} / \sim$$

where the union is over all vertex-weighted stable graphs (G, w) of genus g . Each cell $C(G)$ is an open dense subset of $\mathbb{R}_{\geq 0}^{E(G)}$.

6.1. Proof.

Proof of Theorem 3. It suffices to show that when a graph in a cell of $\mathcal{M}_g^{\text{graph}}$ approaches the boundary of its cell, the limiting value of the rank-generating power series agrees with the rank-generating power series of the limiting graph. In an equation,

$$\lim_{t \rightarrow 0} T^+(G_t; u, w) = T^+(\lim_{t \rightarrow 0} G_t; u, w).$$

Approaching the boundary of a cell in $\mathcal{M}_g^{\text{graph}}$ means that the length of an edge is approaching zero. It suffices to check that $\lim_{\alpha \rightarrow 0} \llbracket \alpha \rrbracket_{1+u} = 0$. $\square$

6.2. Tropical curves. The moduli space of metric graphs of fixed genus is not compact. It has a natural compactification in which the extra points at the boundary correspond to tropical curves. Here we use “tropical curve” to refer to a metric graph which possibly has contracted loops, which we think of as “infinitesimally small” loops attached to a vertex. We record the number of

7. FURTHER QUESTIONS

Observation: for a fixed combinatorial graph G , the (i, j) -coefficient of the rank-generating power series $T^+(\Gamma; u, w)$ is a polynomial in the edge lengths.

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APPENDIX: SPECIALIZATIONS OF THE TUTTE POLYNOMIAL

7.1. Constants. For a graph $G = (V, E)$, the Tutte polynomial has the following specializations to graph invariants when evaluated at particular integer points.

- $T^+(G; 1, 1)$ = the number of edge subsets; $T_G(2, 2) = 2^{\#E}$.
- $T^+(G; 0, 0)$ = the number of spanning trees.
- $T^+(G; 0, 1)$ = the number of connected spanning subgraphs.
- $T^+(G; 1, 0)$ = the number of acyclic spanning subgraphs.
- $T^+(G; -1, 1)$ = the number of totally cyclic orientations.
- $T^+(G; 1, -1)$ = the number of acyclic orientations.
- $T^+(G; -k, -1) = (\pm 1/k) \cdot$ the number of k -colorings.

For a metric graph $\Gamma = (G, \ell)$,

$$\begin{aligned} T^+(\Gamma; 1, 1) &= \sum_{A \subseteq E(G)} \prod_{e_i \in A} [\ell(e_i)]_2 = \sum_{A \subseteq E(G)} \prod_{e_i \in A} (2^{\ell(e_i)} - 1). \\ &= \prod_{e_i \in E(G)} (1 + (2^{\ell(e_i)} - 1)) = 2^{\sum_i \ell(e_i)} \end{aligned}$$

- $T^+(\Gamma; 1, 1) = 2^{\text{vol}(\Gamma)}$
- $T^+(\Gamma; 0, 0) = \text{vol}(\text{Jac}(\Gamma))$
- $T^+(\Gamma; 0, 1) = \sum_{k=0}^g \text{vol}(\text{Eff}^k(\Gamma))?$

Example 15. Suppose Γ is the circle graph with edge length a . Then

- the number of spanning trees is $T^+(\Gamma; 0, 0) = a$
- the number of connected spanning subgraphs is $T^+(\Gamma; 0, 1) = a + 1$;
- the number of acyclic spanning subgraphs is $T^+(\Gamma; 1, 0) = 2^a - 1$;
- the number of totally acyclic orientations is $T^+(\Gamma; 1, -1) = 2^a - 2$;
- the number of totally cyclic orientations is $T^+(\Gamma; -1, 1) = 2$;

Example 16. Suppose Γ is the theta graph with edge lengths a , b and c ,

$$\text{ev}_{(2,2)}(\Gamma) = 2^{a+b+c}.$$

Number of spanning trees:

$$\text{ev}_{(1,1)}(\Gamma) = T^+(\Gamma; 0, 0) = ab + ac + bc.$$

Number of connected spanning subgraphs:

$$\text{ev}_{(1,2)}(\Gamma) = T^+(\Gamma; 0, 1) = 1 + (a + b + c) + (ab + ac + bc).$$

Number of acyclic spanning subgraphs:

$$\text{ev}_{(2,1)}(\Gamma) = T^+(\Gamma; 1, 0) = 2^{a+b+c} - 2^a - 2^b - 2^c + 2.$$

Number of totally cyclic orientations:

$$\text{ev}_{(0,2)}(\Gamma) = T^+(\Gamma; -1, 1) = 1 + 3 + 3 - 1 = 6.$$

Number of totally acyclic orientations:

$$\text{ev}_{(2,0)}(\Gamma) = T^+(\Gamma; 1, -1) = 2^{a+b+c} - 2(2^a + 2^b + 2^c) + 6.$$

Example 17. For the theta graph, we have

$$\begin{aligned} T^+(\Gamma; u, w) &= w^2 + (\llbracket a \rrbracket_{1+u} + \llbracket b \rrbracket_{1+u} + \llbracket c \rrbracket_{1+u})w \\ &\quad + (\llbracket a \rrbracket_{1+u} \llbracket b \rrbracket_{1+u} + \llbracket a \rrbracket_{1+u} \llbracket c \rrbracket_{1+u} + \llbracket b \rrbracket_{1+u} \llbracket c \rrbracket_{1+u}) \\ &\quad + (\llbracket a \rrbracket_{1+u} \llbracket b \rrbracket_{1+u} \llbracket c \rrbracket_{1+u})u \end{aligned}$$

The Tutte constant coefficient is

$$\text{coeff}(0, 0; \Gamma) = ab + ac + bc.$$

The Tutte coefficient of w is

$$\text{coeff}(0, 1; \Gamma) = a + b + c.$$

The Tutte coefficient of $u^0 w^k$ is

$$\text{coeff}(0, 2; \Gamma) = 1.$$

$$\text{coeff}(0, k; \Gamma) =$$

The Tutte coefficient of u is

$$\begin{aligned} \text{coeff}(1, 0; \Gamma) &= a \binom{b}{2} + \binom{a}{2} b + \cdots + abc = abc + \frac{1}{2} \sum a^2 b - \sum ab \\ \text{coeff}(2, 0; \Gamma) &= a \binom{b}{3} + \binom{a}{2} \binom{b}{2} + \binom{a}{3} b + \cdots + ab \binom{c}{2} + a \binom{b}{2} c + \binom{a}{2} bc \\ &= \frac{1}{6} \sum ab^3 - \frac{1}{2} \sum ab^2 + \frac{2}{3} \sum ab + \frac{1}{4} \sum a^2 b^2 - \frac{1}{4} \sum ab^2 + \frac{1}{4} ab + \frac{1}{2} abc(a + b + c) - \frac{3}{2} abc \\ \text{coeff}(k, 1; \Gamma) &= \binom{a}{k+1} + \binom{b}{k+1} + \binom{c}{k+1}. \end{aligned}$$

Note that at $w = 0$, we have

$$\begin{aligned} T^+(G; u, 0) &= \llbracket a \rrbracket_{1+u} \llbracket b + c \rrbracket_{1+u} + \llbracket b \rrbracket_{1+u} \llbracket c \rrbracket_{1+u} \\ &= \sum_{k \geq 0} \binom{a}{k+1} u^k \sum_{k \geq 0} \binom{b+c}{k+1} u^k + \sum_{k \geq 0} \binom{b}{k+1} u^k \sum_{k \geq 0} \binom{c}{k+1} u^k \\ &= \left(a + \binom{a}{2} u + \cdots \right) \left(b + c + \binom{b+c}{2} u + \cdots \right) + \left(b + \binom{b}{2} u + \cdots \right) \left(c + \binom{c}{2} u + \cdots \right) \\ &= (ab + ac + bc) + \left(a \binom{b+c}{2} + \binom{a}{2} (b+c) + b \binom{c}{2} + \binom{b}{2} c \right) u + \cdots \\ &= (ab + ac + bc) + \frac{1}{2} (a(b+c)(a+b+c-2) + bc(b+c-2)) u + \cdots \\ &= (ab + ac + bc) + \frac{1}{2} (a^2 b + a^2 c + ab^2 + ac^2 + b^2 c + bc^2 + 2abc - 2ab - 2ac - 2bc) u + \cdots \\ &= (ab + ac + bc) + \left(abc + \frac{1}{2} (a^2 b + ab^2 + a^2 c + ac^2 + b^2 c + bc^2) - ab - ac - bc \right) u + \cdots \end{aligned}$$

$$\begin{aligned} T^+(G; u, 0) &= \llbracket a \rrbracket_{1+u} \llbracket b + c \rrbracket_{1+u} + \llbracket b \rrbracket_{1+u} \llbracket c \rrbracket_{1+u} \\ &= \llbracket a \rrbracket_{1+u} (\llbracket b \rrbracket_{1+u} + \llbracket c \rrbracket_{1+u} + u \llbracket b \rrbracket_{1+u} \llbracket c \rrbracket_{1+u}) + \llbracket b \rrbracket_{1+u} \llbracket c \rrbracket_{1+u} \\ &= [a][b] + [a][c] + [b][c] + u[a][b][c]. \end{aligned}$$

7.2. Chromatic polynomial. At $y = 0$ we obtain the chromatic polynomial of a (connected) graph:

$$\chi(G; \lambda) = (-1)^{\#V} (-\lambda) T(G; 1 - \lambda, 0)$$

For a metric graph,

$$\chi(\Gamma; \lambda) = (-\lambda) T^+(\Gamma; -\lambda, -1) = \sum_{A \subseteq E} (-\lambda)^{h_0(\Gamma \setminus A)} (-1)^{h_1(\Gamma \setminus A)} \prod_{e \in A} \llbracket \ell(e) \rrbracket_{1-\lambda}.$$

7.3. Flow polynomial. At $x = 0$ we obtain the flow polynomial of a graph:

$$F(G; \lambda) = (-1)^{h_1(G)} T(G; 0, 1 - \lambda)$$

For a metric graph,

$$\begin{aligned} F(\Gamma; \lambda) &= (-1)^{h_1(\Gamma)} T^+(\Gamma; -1, -\lambda) = \sum_{A \subseteq E} (-1)^{h_0(\Gamma \setminus A)-1} (-\lambda)^{h_1(\Gamma \setminus A)} \prod_{e \in A} \llbracket \ell(e) \rrbracket_0 \\ &= \sum_{A \subseteq E} (-1)^{\chi(\Gamma \setminus A)} \lambda^{h_1(\Gamma \setminus A)} \end{aligned}$$

Conclusion: (positive) edge lengths don't change the flow polynomial.

7.4. Reliability polynomial. The reliability polynomial of a graph satisfies

$$R(G; p) = (1 - p)^{\#V - h_0(G)} p^{h_1(G)} T\left(G; 1, \frac{1}{p}\right)$$

For a metric graph,

$$\begin{aligned} R(\Gamma; p) &= (1 - p)^{\infty} p^{h_1(\Gamma)} T^+\left(\Gamma; 0, \frac{1 - p}{p}\right) \\ &= (1 - p)^{\infty} p^{h_1(\Gamma)} \sum_{\substack{A \subseteq E \\ \Gamma \setminus A \text{ connected}}} \left(\frac{1 - p}{p}\right)^{h_1(\Gamma \setminus A)} \prod_{e \in A} \llbracket \ell(e) \rrbracket_1 \\ &= (1 - p)^{\infty} \sum_{\substack{A \subseteq E \\ \Gamma \setminus A \text{ connected}}} p^{\#A} (1 - p)^{h_1(\Gamma \setminus A) - \#A} \prod_{e \in A} \ell(e) \end{aligned}$$

7.5. Potts model polynomial. Following Sokal [17, Section 2.5]

The (modified) Potts model polynomial, or cluster-generating function, $\tilde{Z}(G; q, v)$ is

$$\tilde{Z}(G; q, v) = \sum_{A \subseteq E} q^{h_0(G|A) - |V|v|A|} = \sum_{A \subseteq E} q^{h_1(G|A)} (v/q)^{|A|}$$

In terms of the Tutte polynomial, this is

$$\tilde{Z}(G; q, v) = (q/v)^{h_0(G)} (v/q)^{|V|} T(G; 1 + \frac{q}{v}, 1 + v) = (v/q)^{|V| - h_0(G)} T^+(G; \frac{q}{v}, v).$$

$$\begin{aligned} Z(\Gamma; q, v) &= (v/q)^{\infty} T^+(\Gamma; q/v, v) \\ &= (v/q)^{\infty} \sum_{A \subseteq E} (q/v)^{h_0(\Gamma \setminus A) - 1} v^{h_1(\Gamma \setminus A)} \prod_{e \in A} \llbracket \ell(e) \rrbracket_{1+q/v} \end{aligned}$$

APPENDIX: MISCELLANEOUS

7.6. **q-analogs.** Note that the q -analog satisfies the following properties

- (1) (Varying the edge length) If $q_0 > 0$ is fixed and $q_0 \neq 1$, the map

$$\ell \mapsto [\ell]_{q_0} = \frac{q_0^\ell - 1}{q_0 - 1}$$

defines a continuous function from $\mathbb{R}$ to $\mathbb{R}$, which sends $1 \mapsto 1$ and $0 \mapsto 0$.

If $q_0 = 1$, we use the convention that $[\ell]_1 = \ell$.

If $q_0 = 0$, we have $[\ell]_0 = 1$ for any $\ell > 0$.

- (2) (Varying the formal q -parameter) If $\ell_0 \geq 0$ is fixed and $q > 0$, the map

$$q \mapsto [\ell_0]_q = \frac{q^{\ell_0} - 1}{q - 1}$$

defines a continuous function from $\mathbb{R}_{>0} \setminus \{1\}$ to $\mathbb{R}$, which satisfies

$$\lim_{q \rightarrow 0^+} [\ell_0]_q = \lim_{q \rightarrow 0^+} \frac{q^{\ell_0} - 1}{q - 1} = \begin{cases} 1 & \text{if } \ell_0 > 0 \\ 0 & \text{if } \ell_0 = 0 \\ -\infty & \text{if } \ell_0 < 0. \end{cases}$$

and has a continuous extension to $\mathbb{R}_{>0} \rightarrow \mathbb{R}$ that sends $1 \mapsto \ell_0$.

- (3) In particular, for $\ell, q > 0$ we have

$$[\ell]_0 = 1 \quad \text{and} \quad [0]_q = 0.$$

$$\lim_{\ell \rightarrow 0} [\ell]_0 = 1 \quad \text{and} \quad \lim_{q \rightarrow 0} [0]_q = 0.$$

Considering $[\alpha]_{1+u}$ as a power series in u and α :

$$\begin{aligned} \llbracket \alpha \rrbracket_{1+u} &= \sum_{k \geq 0} \binom{\alpha}{k+1} u^k = \sum_{k \geq 0} \frac{1}{(k+1)!} \alpha^{k+1} u^k \\ &= \sum_{k \geq 0} \left(\sum_{j \geq 0} \frac{s(k+1, j)}{(k+1)!} \alpha^j \right) u^k \\ &= (\alpha) + \left(-\frac{1}{2}\alpha + \frac{1}{2}\alpha^2\right)u + \left(\frac{1}{3}\alpha - \frac{1}{2}\alpha^2 + \frac{1}{6}\alpha^3\right)u^2 + \dots \end{aligned}$$

where $x^{\underline{k}} = x(x-1)(x-2)\cdots(x-k+1)$ denotes the falling factorial and $s(k, j)$ denotes the Stirling number of the first kind.

As a power series in α :

$$\begin{aligned} \llbracket \alpha \rrbracket_{1+u} &= \frac{(1+u)^\alpha - 1}{u} = \frac{\exp(\alpha \log(1+u)) - 1}{u} \\ &= \frac{1}{u} \left(-1 + \sum_{j \geq 0} \frac{\log(1+u)^j}{j!} \alpha^j \right) \\ &= \sum_{j \geq 1} \frac{\log(1+u)^j}{j! u} \alpha^j. \end{aligned}$$

Maybe remove
this theorem

7.7. Rank-generating power series coefficients.

Theorem 18 (Continuity of Tutte coefficient). *For fixed indices $i, j \geq 0$, let $\text{coeff}(i, j; \Gamma)$ denote the coefficient of $u^i w^j$ in the power series expansion of $T^+(\Gamma; u, w)$. Then the function $\text{coeff}(i, j)$ defines a locally-polynomial function $\mathcal{M}_g^{\text{graph}} \rightarrow \mathbb{R}$.*

extra assumptions
needed?

7.8. Rank-generating power series as real-valued function. Given real parameters x, y with $x > 0$, let

$$(8) \quad T(\Gamma; x, y) = \sum_{A \subseteq E(G)} \left(\prod_{e \in A} [\ell(e)]_x \right) (x-1)^{h_0(G \setminus A)-1} (y-1)^{h_1(G \setminus A)}$$

where the notation $[\alpha]_x$ for real $\alpha, x > 0$ means

$$[\alpha]_x = \frac{x^\alpha - 1}{x - 1} \quad \text{if } x \neq 1, \quad [\alpha]_1 = \alpha.$$

For a fixed metric graph Γ , the expression (8) defines a function $\mathbb{R}_{>0} \times \mathbb{R} \rightarrow \mathbb{R}$ by associating $(x, y) \mapsto T(\Gamma; x, y)$. This function is generally not a polynomial in x ; moreover, it does not admit a formal power series expansion in x if any edge length $\ell(e_i)$ is non-integral.

It is straightforward to verify that the rank-generating power series $T^+(\Gamma; u, w)$ converges to a real value when $|u| < 1$. For a generic choice of edge lengths, the radius of convergence in u is equal to 1.

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